Assignment #1

STAT 679: Statistical Methods for Spatial Lattice Data–Spring 2026, UW-Madison

1. **(20 pts)** Consider the following data, which represent the locations of bowhead whales. Basically, the locations of a number of bowhead whales (each of which had been tagged with a radio transmitter), were noted within a region in the North Pacific. Each quadrat below represents a region 20 km by 20 km.

$$\begin{array}{cccccc} 1 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 2 & 2 & 1 & 2 & 0 & 3 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{array}$$

   (a) Label any non-zero quadrat "black" and any zero quadrat "white." Using the R function `joincount.multi()`, conduct a z-test for H_0 : CSR vs. H_A : not CSR, based on BB, BW, and WW joins for these data. Assume first-order ("rook's case") neighborhood structure. Use binary spatial weights (`style="B"`). Report the z-statistics and p-values for each test. You do not need to use the Monte Carlo tests here.

   (b) Do you reject the null hypothesis of "randomness"? Interpret the results from (a).

   (c) What can you say about the pattern in the data?

2. **(10 pts)** Repeat the exercise above using a second-order ("queen's case") neighborhood structure. Comment on the differences you observe.

3. **(5 pts)** By hand, list out the first-order neighbors for each cell in mosaic pattern II. Compare your answers to the `str(egII.nb)`. How many did you get? Repeat the same exercise for `type="queen"` (second-order neighbors). Submit your hand calculations (image).